LAB: 9 and 10

WAP to exchange values between two variables.

Title: To Swap Variables.

```
 .MODEL SMALL
  .STACK 100H
  .DATA
     NUM1 DW 1234H
     NUM2 DW 4567H
   .CODE
       MAIN PROC
        MOV AX, @DATA
        MOV DS, AX
        MOV AX, NUM1            ;AX=NUM1
        MOV BX, NUM2            ;BX=NUM2
        MOV NUM2, AX            ;NUM2=AX
        MOV NUM1, BX            ;NUM1=BX
        MOV AH, 4CH
        INT 21H
      MAIN ENDP
     END MAIN
```

OUTPUT:

NUM1= 4567

NUM2=1234

Program to take input from user and display the result.

```
.model small
.stack 100h
.data
.code
main proc
   mov ah, 1   ;1 is used for input from keyboard
```

```
    int 21h
    mov dl, al  ; content of accumulator is moved into dl register
    mov ah, 2   ; used for output
    int 21h
    main endp
   end main
```

Program to print single and multiple character on screen using assembly language

```
.model small
.stack 100h
.data
.code
main proc
    mov dl, "M"
    mov ah, 2
    int 21h
    mov dl, "U"
    mov ah, 2
    int 21h
     mov dl, "K"
    mov ah, 2
    int 21h
    mov dl, "E"
    mov ah, 2
    int 21h
     mov dl, "S"
    mov ah, 2
    int 21h
    mov dl, "H"
    mov ah, 2
    int 21h
```

```
 mov ah, 4ch
int 21h
main endp
end main
```

WAP to Print A to Z in small and capital letter.

```
.model small
.stack 100h
.data
.code
main proc
    mov cx, 26          ;Set counter
    mov dx, 65          ; ASCII CODE of a=97 A=65 and Z =90  z=122  for Z to A use 90
     test:
    mov ah,2
    int 21h
    inc dx ;      ; To Print z to a use dec
    loop test
    MAIN ENDP
    END MAIN
```

Program to convert Capital letter to small and small letter to capital in assembly language

```
.model small
.stack 100h
.data
.code
main proc
    mov ah,1        ; input  a=97, A=65; 97-65=32
    int 21h
     mov dl,al       ;ascii value of input is moved into register dl
    add dl,32       ; To convert in small letter ; To convert into capital letter subtract 32
    mov ah,2        ; to display result character by character
```

```
    int 21h
    mov ah, 4ch
    int 21h
    main endp
 end main
```

Program to print two message declared in data segment with space character and in new line on screen in assembly language.

```
.model small
.stack 100h
.data
var1 db "HELLO$"
var2 db " WORLD$"
;var2 db 10,13, "WORLD$"        ;To print in new line
.code
main proc
    mov ax, @data       ;  address of data segment is moved into register AX
    mov ds, ax          ;address of ax register is moved into ds
    mov dx, offset var1 ; address of var1 is moved to dx register
    mov ah, 9           ;To show the string
    int 21h
     lea dx, var2 ; address of var1 is moved to dx register
    mov ah, 9           ;To show the string
    int 21h
    mov ah, 4ch
    int 21h
    main endp
 end main
```

Assembly language program to print number from 0-9 or 9-0.

```
.model small
.stack 100h
.data
.code
main proc
   mov cx, 10 ;set counter to 10
   mov dx, 48 ;ascii code of zero(0); to print 9 to 0 put 57 in place of 48
    test:
   mov ah, 2  ; print the number
    int 21h
   inc dx       ; dec is used in place of inc to print from 9 to 0
   loop test
    mov ah, 4ch
   int 21h
   main endp
 end main
```

WAP to compare two number whether it equal or not.

```
org 100h
.model small
.stack 100h
.data
   msg1 db 10,13,"Enter First Number:$"
   msg2 db 10,13,"Enter Second Number:$"
   msg3 db 10,13,"Number are Equal..$"
   msg4 db 10,13,"Number are not Equal..$"
.code
main proc
    mov ax,@data          ;Memory heap created to store the data address present in data segment in ax.
    mov dx,ax
```

```
 mov dx,offset msg1
mov ah,9              ;to show message1
int 21h
 mov ah,1             ; to take input from keyboard
int 21h
mov cl,al            ; store first number in cl
 mov dx, offset msg2
mov ah,9             ;to show message2
int 21h
 mov ah, 1             ; to take input from keyboard
int 21h
mov dl,al            ; store Second number in dl
cmp dl,cl            ;Compare first and second number
  je label1          ; jump if equal to label1
 mov dx, offset msg4    ; if number are not equal print message4
mov ah,9
int 21h
mov ah,4ch
int 21h
label1:
mov dx,offset msg3
mov ah,9
int 21h
 mov ah,4ch
int 21h
main endp
end main
ret
```